

Tat-Chuyen Mai

AI Researcher — Specializing in Computer Vision and Deep Learning

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Career Objective

Dedicated AI researcher with **three years** of experience in developing & optimizing deep learning architectures for image segmentation, human pose estimation, **first authored of 4 scientific papers**. Strong background in computer vision, with hands-on experience in coding, training, and fine-tuning deep learning models using Python on NVIDIA RTX 3090 GPUs. Eager to apply vision-based AI to automation, building reliable models that bridge perception and intelligent decision-making.

Education

Hanoi University of Science and Technology (HUST)

Sep 2021 – Sep 2025

Bachelor of Advanced Program in Biomedical Engineering

- GPA: **3.72/4.00**
- Academic Excellence Scholarship [↗](#)
- Best Graduation Project [↗](#)
- 2nd Prize (School) [↗](#) & 3rd Prize (University) [↗](#) – HUST Annual Student Research Conference 2024–2025
- Related subjects: Programming Languages, Probability and Statistics, Machine learning, Image processing

Publications

• Inception Vision Mamba Unet for Tumor Segmentation on Breast Ultrasound Images

Tat-Chuyen Mai, Viet-Dung Nguyen* - [Springer Nature](#) [↗](#)

Developed the Vision Mamba U-Net architecture integrating multi-scale feature learning from Inception modules, achieving a 3% improvement in segmentation accuracy on breast ultrasound datasets.

• Motion Analysis & Performance Monitoring System for Squat Exercise by Simple Estimation

Mai Tat Chuyen, Trinh Khanh Ly, Ong Thi Huyen Trang, Nguyen Trung Phu, Nguyen Thanh Hai, Nguyen Thuy Anh, Tran Duc Tan, Nguyen Phan Kien* - [ICISN 2025](#) [↗](#)

Proposed a lightweight algorithm using MediaPipe for real-time human pose extraction and joint-angle estimation in motion analysis applications.

• Multimodal Deep Learning for Breast Ultrasound Lesion Classification

Tat-Chuyen Mai, Khanh-Ly Trinh, Huyen-Trang Ong, Hai-Long Nguyen, Viet-Dung Nguyen* - [ICHST 2025](#) [↗](#)

Proposed a multimodal deep learning framework integrating image and metadata representations to enhance classification of breast ultrasound lesions.

• Multichannel Learning Framework for Enhanced ECG Signal Classification

Tat-Chuyen Mai, Diem-Anh Nguyen Thi, Quang-Huy Hoang, Viet-Huong Pham Thi, Anh-Vu Tran* - [Iranian Journal of Science and Technology \(Q2\)](#) [↗](#)

Proposed a multimodal deep learning framework that integrates spectrogram images of ECG signals with wavelet-transformed filtered raw signals.

Experience

Research Assistant

Biomedical Signal and Image Analysis Laboratory (BSI), HUST

Sep 2022 – Sep 2025

- Conducted research on **AI-based breast ultrasound image analysis**, focusing on tumor segmentation and lesion classification.

- Explored multimodal & multitask learning techniques to enhance learning performance.
- Trained large-scale vision models on NVIDIA RTX 3090 GPU with optimized CUDA pipelines for efficient performance.
- Deployed a breast cancer segmentation model as a web-based demo, integrating a Python-based backend inference API with a web frontend interface.
- Deployed ECG classification model on STM32 MCU for real-time arrhythmia detection
- *Tools: PyTorch, Python, Deep Learning, Image Segmentation, Model Optimization, Scientific Writing.*

Software Engineer Intern

Motion Analysis Lab, Vinmec Times City International Hospital

Sep 2024 – Apr 2025

- Developed a Python-based **motion analysis and visualization system** integrating data from multiple camera-based measurement tools; **currently used by doctors (Vinmec)** for clinical evaluation.
- **Project code** and description details available at: [Project Page](#) 
- *Tools: Python, Data Processing, Motion Analysis, GUI Development.*

Skills

- **Programming:** Python, C/C++
- **AI & ML Frameworks:** PyTorch, TensorFlow, Scikit-learn
- **Data Processing & Visualization:** NumPy, Pandas, Matplotlib
- **Tools & Systems:** Git, VS Code, Jupyter, LaTeX, Linux, Windows, CUDA
- **Language:** English (VSTEP B2)

Computer Vision and Deep Learning Expertise

- **Deep Learning:** CNNs, UNet, Transformer, Mamba, RNNs, GNNs
- **Image Processing:** Edge Detection, Denoising, Feature Extraction, Filtering, and Enhancement
- **Generative Vision Models:** Diffusion Models (Theory and fine-tuned)

Online Certifications

- [Google Advanced Data Analytics Professional Certificate](#)  – [Coursera](#) 
- [Machine Learning Specialization](#)  – [Coursera](#) 
- [AI for Medicine Specialization](#)  – [Coursera](#) 

Professional Activities

- **Reviewer**, IEEE CIBCB 2025 – 22nd IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology.

Referees

Dr. Viet-Dung Nguyen, School of Electrical and Electronics Engineering, HUST

- [Google Scholar](#)  • Email: dung.nguyenviet1@hust.edu.vn